Chapter 5: Convex Optimization

One concept, one exercise per slide — practice as you go

Concept: Convex Sets

- A set is convex if the straight line between any two of its points stays inside the set
- Formally: for u, v in C and t in [0,1], the point t·u + (1-t)·v is also in C
- A disk is convex; a crescent (moon) shape is not

✏️ **Exercise: Is the set of all points with x ≥ 0 (a half-plane) convex? Is a donut shape (an annulus) convex?**

Concept: Convex Functions (Jensen's Inequality)

- f is convex if $f(t \cdot u + (1-t) \cdot v) \leq t \cdot f(u) + (1-t) \cdot f(v)$ for all u, v and t in [0,1]
- In words: the function's value on the line segment is never above the straight line connecting the endpoints
- Equivalent second-order test: the Hessian (second derivative matrix) is positive semidefinite everywhere

✏️ **Exercise: Using Jensen's inequality with $u=0$, $v=4$, $t=0.5$, check whether $f(x)=x^2$ is convex at this pair of points.**

Concept: Local Minima Are Global Minima (For Convex Functions)

- The single most important theorem in this chapter
- If f is convex, ANY local minimum is automatically the GLOBAL minimum
- This is why gradient descent on a convex loss is trustworthy, not just hopeful
- For non-convex functions (most neural networks), this guarantee simply does not exist

✏️ **Exercise: Explain, in one sentence, why this theorem is the mathematical reason linear regression's normal equations and gradient descent always agree on the same answer.**

Concept: Strict and Strong Convexity

- Strict convexity: Jensen's inequality is STRICT — guarantees at most one global minimizer
- Strong convexity ($\mu > 0$): quantifies HOW sharply the function curves upward from its minimum
- L2 regularization (weight_decay) adds λI to the Hessian, guaranteeing strong convexity
- This is why ridge regression always has a unique solution, even with collinear features

✏️ **Exercise: Plain (unregularized) linear regression on perfectly collinear features has infinitely many equally-good solutions. Why does adding an L2 penalty fix this?**

Concept: Gradient Descent's Safe Step Size

- L-smoothness: the gradient cannot change faster than rate L (a Lipschitz bound)
- The descent lemma proves: any step size $\eta < 2/L$ guarantees the loss decreases every step
- Too large a learning rate ($\eta > 2/L$) can cause the loss to increase, even on a convex function

✏️ **Exercise: A loss function has smoothness constant $L=4$. Which of these learning rates is guaranteed safe: $\eta=0.3$, $\eta=0.5$, $\eta=0.6$?**

Concept: The Condition Number and Convergence Speed

- $\kappa = \lambda_{max} / \lambda_{min}$ — the ratio of the Hessian's largest to smallest eigenvalue
- $\kappa \approx 1$: a round bowl — gradient descent converges in very few steps
- $\kappa \gg 1$: an elongated bowl — gradient descent zig-zags and converges slowly
- Feature scaling reshapes an elongated bowl back toward round, directly speeding up training

✏️ **Exercise: Two features are on wildly different scales: one ranges 0-1, the other ranges 0-100,000. Predict, qualitatively, whether training will converge quickly or slowly before any fix is applied — and name the one-line fix.**

Concept: Newton's Method — Using Curvature

- Gradient descent only uses the SLOPE (first derivative) at the current point
- Newton's method also uses the CURVATURE (the Hessian) to jump straight to the minimum
- On a true quadratic function, Newton's method converges in exactly ONE step, any κ
- Cost: forming and inverting the Hessian is $O(D^3)$ — prohibitive for large D

✏️ **Exercise: Why does modern large-scale machine learning almost universally use gradient descent (or its variants) instead of Newton's method, despite Newton's method needing far fewer steps?**

Concept: Historical Origins — Cauchy to Robbins-Monro

- 1847: Cauchy introduces gradient (steepest) descent for solving astronomical equations
- 1805/1795: Legendre and Gauss solve least squares in closed form — no iteration needed
- 1951: Robbins and Monro formalize stochastic approximation — the ancestor of mini-batch SGD

✏️ **Exercise: Why was gradient descent not necessary at all for solving the earliest least-squares problems, even though it was invented decades later?**

Mastery Check: Are You Ready to Move On?

If you can answer every question below confidently, you have mastered Chapter 5.

1. Define a convex set and a convex function, both geometrically and via Jensen's inequality.
2. State and prove (in outline) why every local minimum of a convex function is global.
3. Explain the difference between strict convexity and strong convexity.
4. Explain why adding L2 regularization guarantees a unique minimizer even under collinearity.
5. Derive the safe learning rate bound $\eta < 2/L$ from the descent lemma.
6. Define the condition number and explain its geometric meaning and effect on convergence speed.
7. Explain why feature scaling speeds up gradient descent in terms of the condition number.
8. Derive why Newton's method converges in exactly one step on a quadratic function.
9. Explain why Newton's method is impractical at the scale of modern neural networks.
10. Implement gradient descent from scratch and empirically verify the $\eta < 2/L$ convergence boundary.